



CTRL FREAKS

SOLUTION REPORT

# ARTHIX

## AI-Powered Hyper-Personalized Banking

*The right product. The right moment. Or no product at all.*

A product and architecture report for an intelligence layer that understands financial context, speaks the customer's language, protects stressed customers, and recommends products only when they create genuine value.

PREPARED BY  
CTRL FREAKS

DATE  
12 September 2026

CORE ENGINE  
Vivek — Recommend / Assist First / Suppress

MODULES  
ArthBodh · JeevanChakra · BhashaSahayak · Sahara ·  
Kavach · Nyay

MVP STACK  
React · Express · SQLite/Postgres · Hybrid AI

LANGUAGES  
Hindi · Gujarati · English (text; voice is future)

This architecture is designed with regulatory readiness and applicable banking/privacy requirements in mind. Production deployment would require formal compliance validation. This document does not claim that the prototype is RBI-compliant or DPDP-certified.

# ARTHIX

AI-Powered Hyper-Personalized Banking

## Team & Selected Problem Statement

### SELECTED PROBLEM STATEMENT

#### AI-Powered Hyper-Personalized Banking for Bharat

Indian banks have built strong digital rails — UPI, net banking, mobile apps, video KYC — yet customers in Tier 2/3/4 towns and rural India still meet generic, confusing journeys. Banks hold rich transactional data but struggle with acquisition cost, drop-offs, and mistimed product offers. ARTHIX addresses this by understanding consented financial context, assisting in vernacular, protecting stressed customers, and recommending a product only when it creates genuine value.

### TEAM



## CTRL FREAKS

Team behind ARTHIX

### TEAM MEMBERS

#### AUMPETHANI

Team Leader

#### DHRUV UPADHYAY

Team Member

#### KRISH TRIVEDI

Team Member

#### ANJALI TRIPATHI

Team Member

## 1. Executive Summary

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Indian digital rails are strong. Bharat experiences are not. Banks hold rich transactional ledgers, yet the same generic offers reach a first-time Gujarati user, a salaried Tier-2 professional, and a household already missing an EMI.

ARTHIX is a customer-first intelligence layer on existing banking channels. It is not a new bank and not a chatbot with banners. It turns consented transactions, EMI, salary, and savings behavior into a living financial profile; guides first-time digital users in Hindi, Gujarati, or English; detects financial stress and unusual activity early; and recommends a loan, insurance, investment, or card only when need, eligibility, suitability, and safety align.

The spine of the product is the **Vivek Engine**: Recommend, Assist First, or Suppress. When a customer is vulnerable, aggressive credit is withheld and assistance is offered instead. Every decision carries reasons, consent checks, and an audit trail.

The architecture is designed with regulatory readiness and applicable banking/privacy requirements in mind.

Production deployment would require formal compliance validation. This document does not claim that the prototype is RBI-compliant or DPDP-certified.

## 2. Problem and Context

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### 2.1 What is broken

Indian banks have UPI, net banking, mobile apps, and video KYC. A large share of customers in Tier 2/3/4 towns and rural India still find those apps confusing, generic, and disconnected from actual financial life.

At the same time, banks face high customer-acquisition cost, drop-offs in loan and product journeys, and weak cross-sell timing. The common cause is a one-size-fits-all digital experience despite rich transactional and behavioral data.

Three failures follow:

1. **Relevance.** Offers ignore life-stage and cash-flow. A salary step-up and a missed EMI receive the same pop-up.
2. **Access.** English-first copy, jargon, and long forms cause first-time digital users to abandon onboarding, KYC, and loan steps.
3. **Conduct.** Stress and fraud are often treated punitively. Customers hide, default, or lose trust.

Banks feel this as wasted campaigns, incomplete journeys, failed cross-sell, rising NPA formation, and grievances.

### 2.2 Why existing approaches fall short

Typical stacks already include generic banners, rule-based campaigns, FAQ chatbots, a separate fraud engine, and collections after delinquency. They do not share one customer financial profile. They have no obligation to *not* sell. Explainability to the customer is weak. Ethics usually sits in a policy PDF, not in a runtime gate.

A vernacular chatbot alone does not solve this. Personalization that always shows a product does not solve this either.

## 3. Solution Overview

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### ARTHIX

**Tagline:** The right product. The right moment. Or no product at all.

**Value:** Financial wellbeing first; products second. Personalization that can say no.

ARTHIX sits on app, net banking, assisted branch, and contact-centre channels as an intelligence layer. One profile drives the dashboard, the assistant, the recommendation card, and the suppress decision.

### Operating loop

1. **Understand** — ingest consented transactions and behavioral signals.
2. **Personalize** — build a living financial profile and need hypotheses.
3. **Assist** — complete hard journeys in the user's language.

- 4. **Protect** — detect stress and fraud; intervene with empathy.
- 5. **Recommend responsibly** — sell only when need, fitness, and safety align.

Objectives

- Personalize from real financial context, not from a name token.
- Reduce drop-off for vernacular first-time users.
- Recommend only suitable, timely, consented products.
- Intervene on stress without predatory credit.
- Verify anomalies with the customer.
- Remain implementable as a hackathon MVP and extendable onto bank infrastructure.

Banking integrations (CBS, UPI, CKYC, notifications, collections) are adapters. In the MVP they are mocked.

4. Core Product Modules

Module 3 (Vivek) is the spine. Modules 1–2 feed it. Modules 4–6 are channels and protections. Module 7 binds every output.

| Module | Name                                                 | Purpose                                    | User problem solved                                         | How it works                                                                            | Benefit                           |
|--------|------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------|
| 1      | <b>ArthBodh</b> — Customer Intelligence              | Build a financial picture from the ledger  | “The app does not know my life”                             | Feature jobs over consented txns: income cadence, spend mix, EMI load, savings residual | Customer is seen as an individual |
| 2      | <b>JeevanChakra</b> — Hyper-personalization          | Turn profile into needs and suitability    | Offers arrive too early, too late, or for the wrong product | Life-stage signals + behavioral segment + need scores                                   | Right family of product, or none  |
| 3      | <b>Vivek Engine</b> — Responsible recommendation     | Decide Recommend / Assist First / Suppress | Generic or harmful pop-ups                                  | Need × eligibility × suitability × stress × leverage gates                              | Useful offer or honest silence    |
| 4      | <b>BhashaSahayak</b> — Conversational Bharat banking | Complete journeys in HI / GU / EN          | English forms and jargon                                    | LLM + intent + grounded tool calls                                                      | First-time users finish steps     |
| 5      | <b>Sahara</b> — Financial wellness                   | Detect stress and help early               | Help arrives after a default flag                           | Change detection + stress band + Assist First                                           | Dignity and a workable next step  |
| 6      | <b>Kavach</b> — Fraud and anomaly                    | Flag unusual behavior                      | Silent drain or sudden freeze                               | Isolation Forest + customer confirm                                                     | Customer stays in the loop        |
| 7      | <b>Nyay Layer</b> — Responsible AI                   | Consent, reasons, fairness, audit          | Invisible decisions, no control                             | Policy gates + reason codes + purpose toggles                                           | Trust and contestability          |

Voice banking, video KYC replacement, live bureau pull, multi-bank aggregation, and rewards gamification are out of the product MVP. They do not strengthen the core loop.

5. User Personas

Rural does not mean “cannot learn.” Salaried does not mean “wants every card.” Stress is not moral failure.

Meena Patel — vernacular first-time digital user

**Profile.** 34, Anand district, shop and farm household income, newly issued debit card, Gujarati preferred, smartphone shared at home.

**Problem.** The app is English. Deposits feel final. She abandons KYC because words such as “nomination” are unclear.

**Behavior.** Completes UPI to known people; avoids unknown menus; trusts the branch more than banners.

**System response.** Gujarati onboarding, short spoken-style copy, chat walks KYC *preparation* (document checklist). No product dump on day one.

**Benefit.** She finishes a journey she previously dropped. The bank gets an activated customer, not a dead download.

### Rahul Deshmukh — salaried Tier-2 customer

**Profile.** 29, Nashik, IT-services salary just stepped up, existing small EMI, Hindi/English mix, digitally confident but offer-fatigued.

**Problem.** Card and loan pop-ups ignore that surplus only stabilized two months ago.

**Behavior.** Checks balance weekly; ignores notifications; would invest if someone explained *why now*.

**System response.** Detect INCOME\_STEP\_UP and SURPLUS\_STABLE. Recommend a small SIP or a card that fits spend mix. Show reasons. Allow dismiss.

**Benefit.** A product that matches a real life-stage, not a spray of twelve offers.

### Kamala Devi — financially stressed customer

**Profile.** 41, peri-urban, two education EMIs, one missed, rising kirana cash-out, Hindi.

**Problem.** A default flag and a gold-loan banner in the same week.

**Behavior.** Avoids the app out of fear; will talk if the tone is practical, not scolding.

**System response.** High stress → suppress unsecured credit → Assist First (EMI calendar, talk-to-bank, expense insight). The fraud path still works if a transaction is odd.

**Benefit.** Dignity and a path. The bank practices early care instead of only collections.

## 6. User Journeys

### Journey A — Personalized recommendation (Rahul)

**Trigger.** Salary credit +20% for two months; EMI ratio below 30%; surplus stable.

**Analysis.** Life-stage plus need INVEST\_SMALL / optional card fit.

**Decision.** Recommend — all Vivek gates pass.

**Interface.** Recommendation card and Why-this.

**Action.** Starts a SIP step or dismisses.

**Outcome.** Relevant product; accept/dismiss is stored as feedback.

### Journey B — Vernacular banking (Meena)

**Trigger.** Opens the app in Gujarati; asks "KYC ma shu joie?"

**Analysis.** Language GU, intent KYC\_HELP; tools fetch the checklist.

**Decision.** Assist — no product.

**Interface.** Chat steps plus dashboard insight "complete KYC when ready."

**Action.** Marks documents ready.

**Outcome.** Lower drop-off; no irrelevant loan.

### Journey C — Financial stress (Kamala)

**Trigger.** Missed EMI and negative surplus slope.

**Analysis.** Stress band High.

**Decision.** Assist First; suppress gold loan and personal loan.

**Interface.** Soft intervention. Recommendations empty with "we are not offering new credit right now."

**Action.** Requests EMI help.

**Outcome.** Empathetic path; predatory credit gap closed.

### Journey D — Fraud verification

**Trigger.** UPI to a new VPA, 6× median, unusual hour.

**Analysis.** Isolation Forest plus rules → Verify.

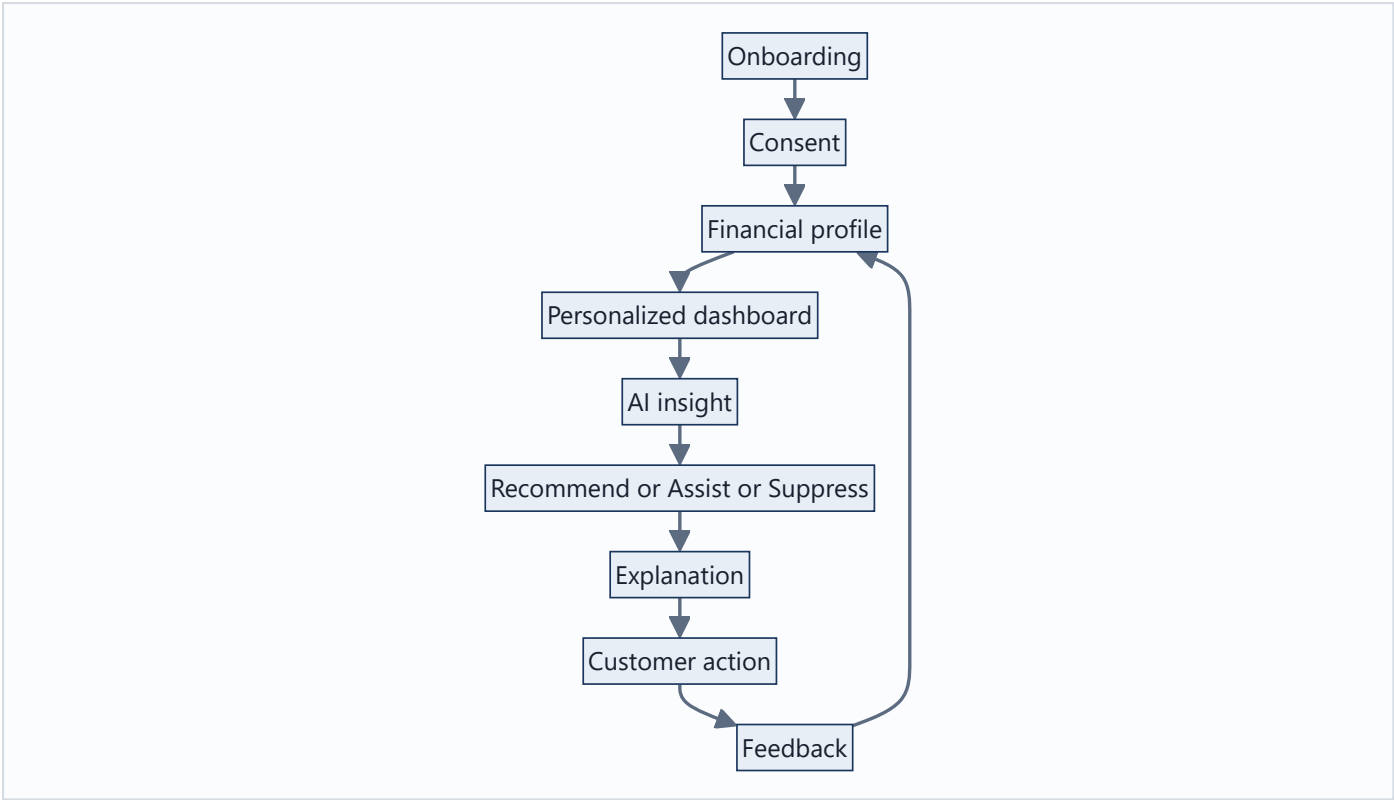
**Decision.** Alert, not a silent freeze.

**Interface.** "Did you send ₹X to Y?"

**Action.** No → escalate. Yes → mark recognized.

**Outcome.** Customer remains in the loop.

DIAGRAM — Customer journey

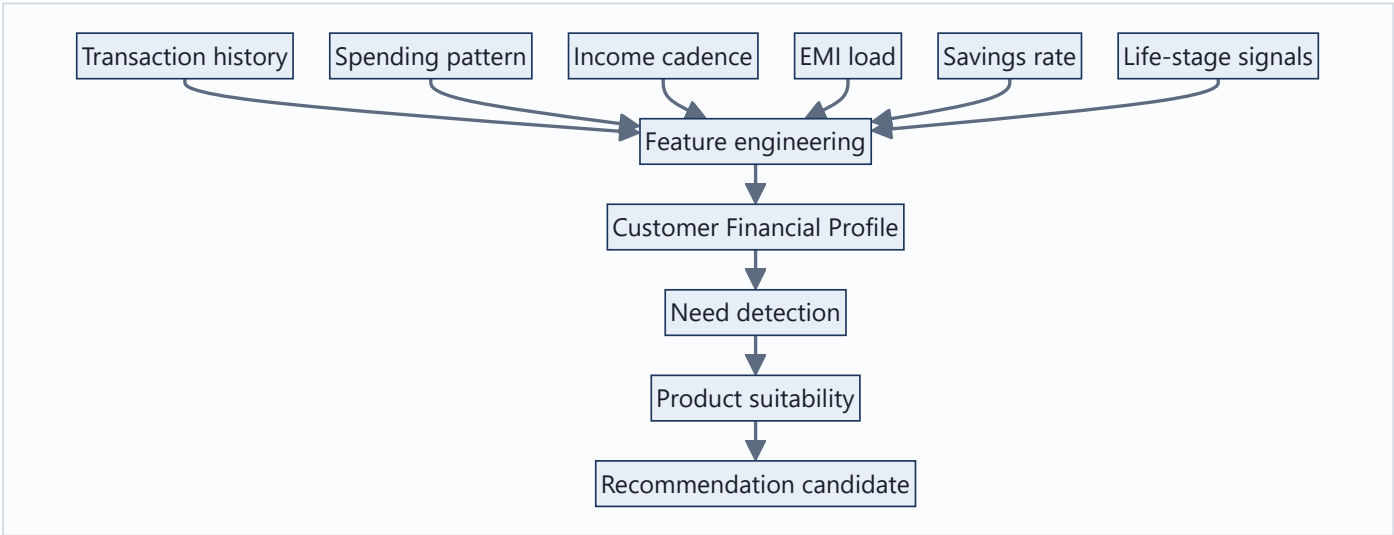


Onboarding and consent come before intelligence. Feedback updates the profile. The same loop serves all three personas.

## 7. Personalization and Decision Intelligence

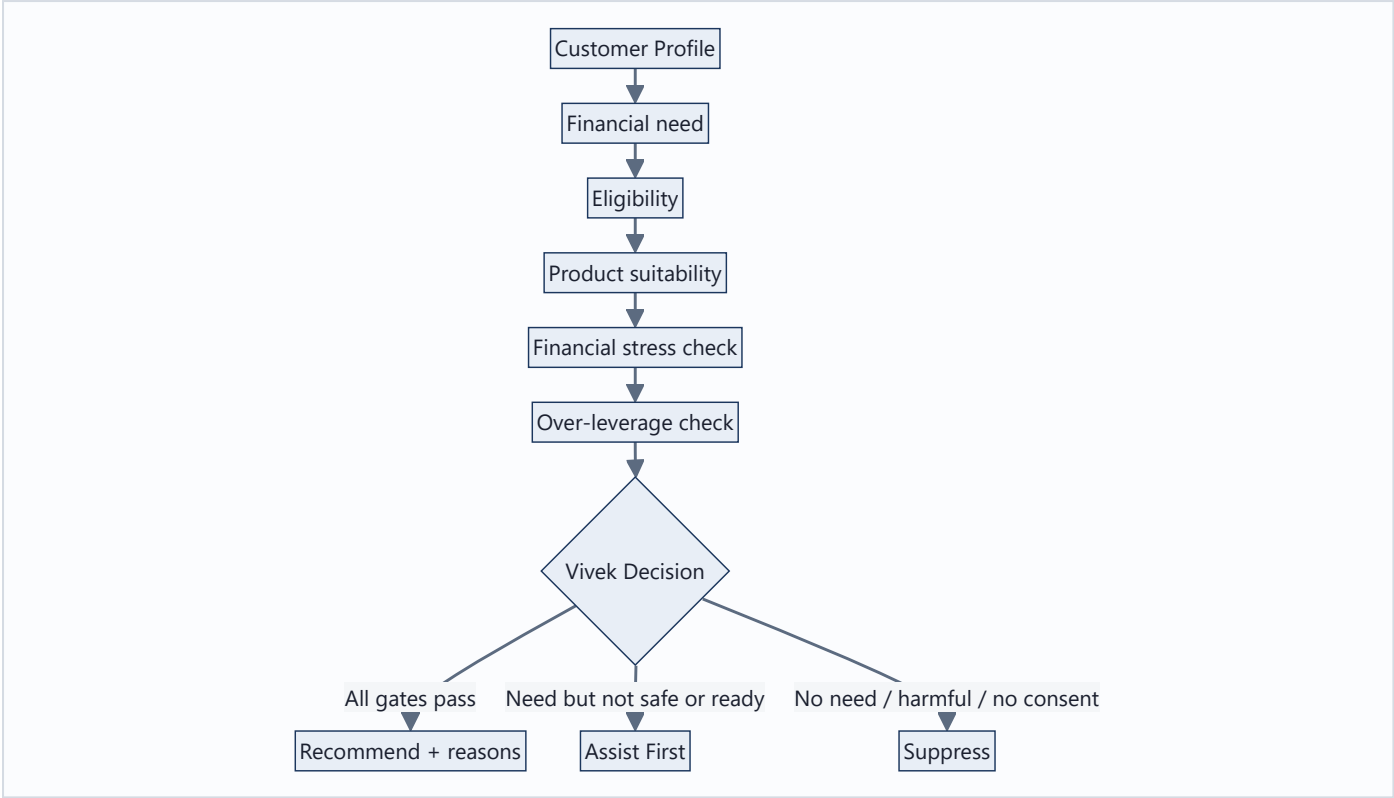
JeevanChakra turns ledger features into need hypotheses. Vivek decides whether a product may appear.

DIAGRAM — Hyper-personalization pipeline



Need, eligibility, and suitability produce a *candidate*. They are not enough to show it.

DIAGRAM — Vivek responsible recommendation



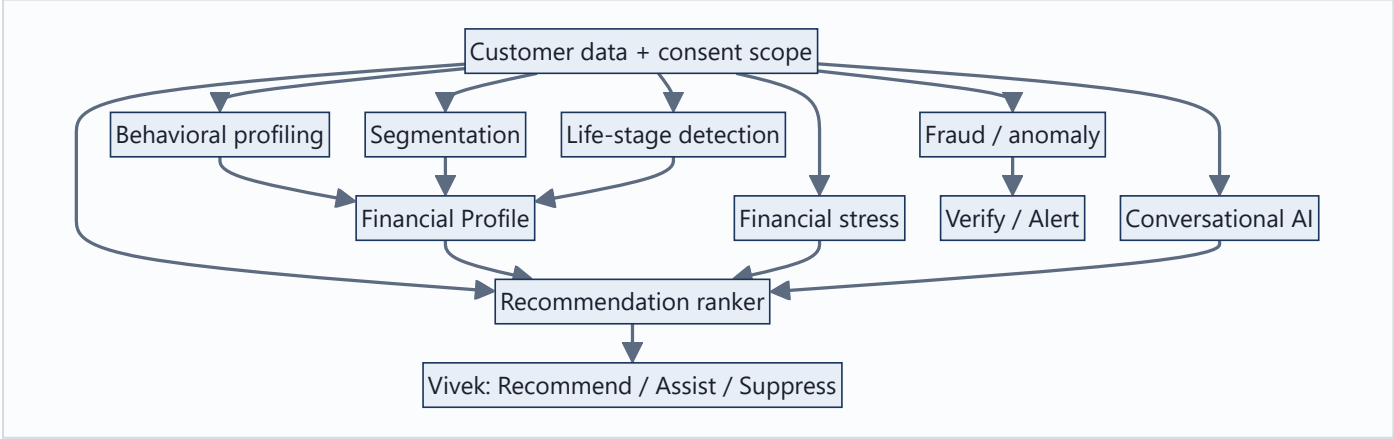
A marketing propensity score cannot override a failed stress or leverage gate. Customer benefit outranks aggressive upselling.

| Gate                | Fail action               |
|---------------------|---------------------------|
| Consent and purpose | No personalization        |
| Genuine need        | Suppress product          |
| Eligibility         | Assist or explain the gap |
| Product suitability | Swap or suppress          |
| Financial stress    | Assist First              |
| Over-leverage       | Suppress credit           |

## 8. AI and ML Architecture

Methods are chosen for explainability and fitness to the data. Deep learning is not used for scoring.

DIAGRAM — AI intelligence



8.1 Behavioral profiling — ArthBodh

|            |                                                                                             |
|------------|---------------------------------------------------------------------------------------------|
| Input      | Consented transactions (amount, date, category, channel, type)                              |
| Features   | Monthly inflows; salary regularity; spend mix; cash vs digital; EMI count; savings residual |
| Method     | Deterministic feature jobs and rolling windows (30 / 90 / 180 days)                         |
| Output     | Financial Profile snapshot                                                                  |
| MVP        | SQL aggregations and a category map on seed ledgers                                         |
| Production | Feature store and merchant MCC enrichment                                                   |

8.2 Segmentation

K-Means (k = 5–7) on scaled profile vectors (income band, EMI ratio, surplus, digital maturity, language). Labels such as SALARIED\_SURPLUS, CASHFLOW\_TIGHT, FIRST\_DIGITAL are a prior only — never the sole reason to sell.

8.3 Life-stage detection

Rule-based detectors on salary, new EMI, rent, education, medical, and travel series. Signals include INCOME\_STEP\_UP, NEW\_CREDIT\_LOAD, SURPLUS\_STABLE, DEPENDENTS\_COST. Life-stage events are sparse; rules stay auditable.

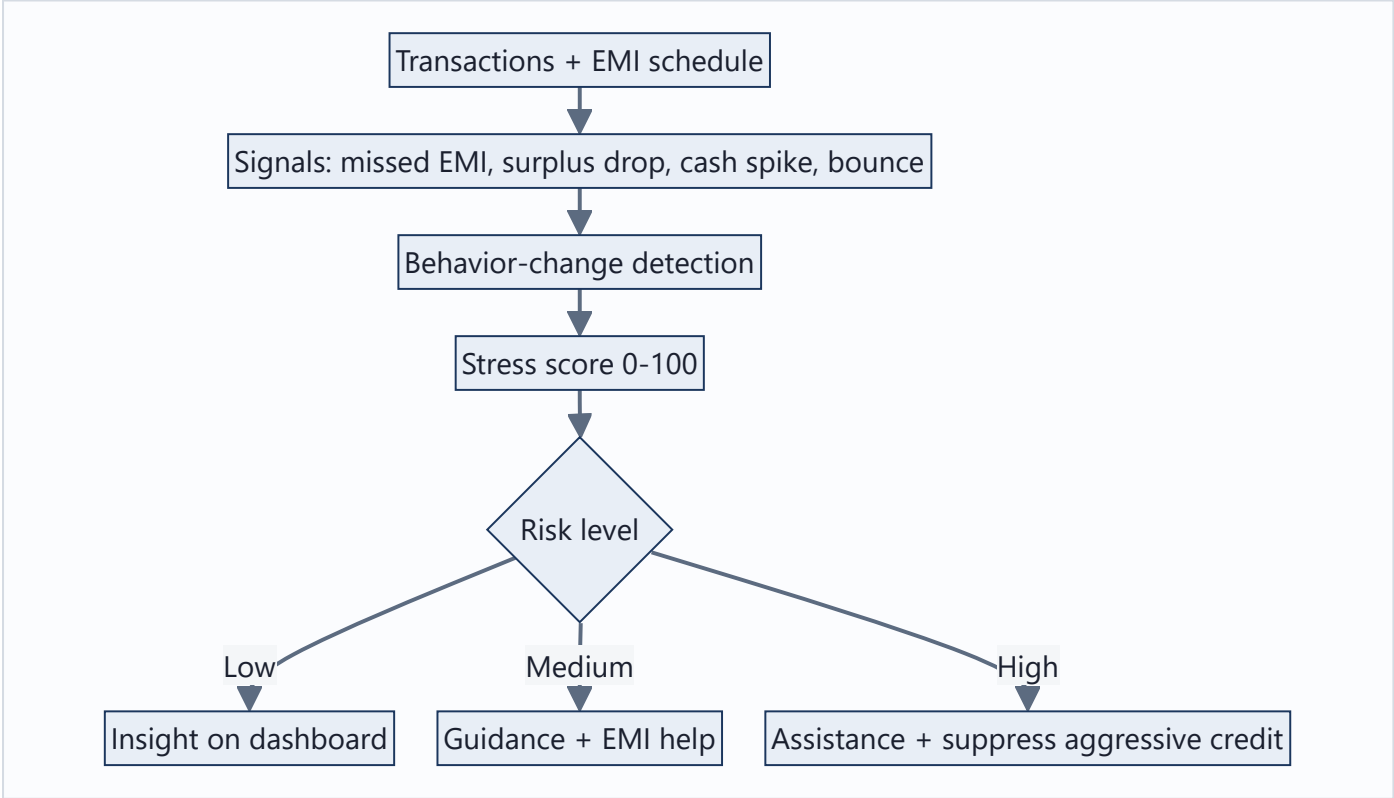
8.4 Need detection and ranking

Hybrid need rules plus a utility ranker. Prototype score:  $0.35 \cdot \text{need} + 0.25 \cdot \text{suit} + 0.20 \cdot \text{elig} + 0.20 \cdot \text{timing}$ . Collaborative filtering is not used — interaction history is thin and opaque. Vivek may discard the entire ranked list.

8.5 Financial stress — Sahara

|          |                                                                             |
|----------|-----------------------------------------------------------------------------|
| Input    | EMI due vs paid, bounce, surplus trend, essential-spend cut, cash-out spike |
| Method   | Rules plus time-series slope                                                |
| Output   | Score 0–100 and band Low / Medium / High                                    |
| Decision | High → Assist and suppress unsecured or aggressive credit                   |

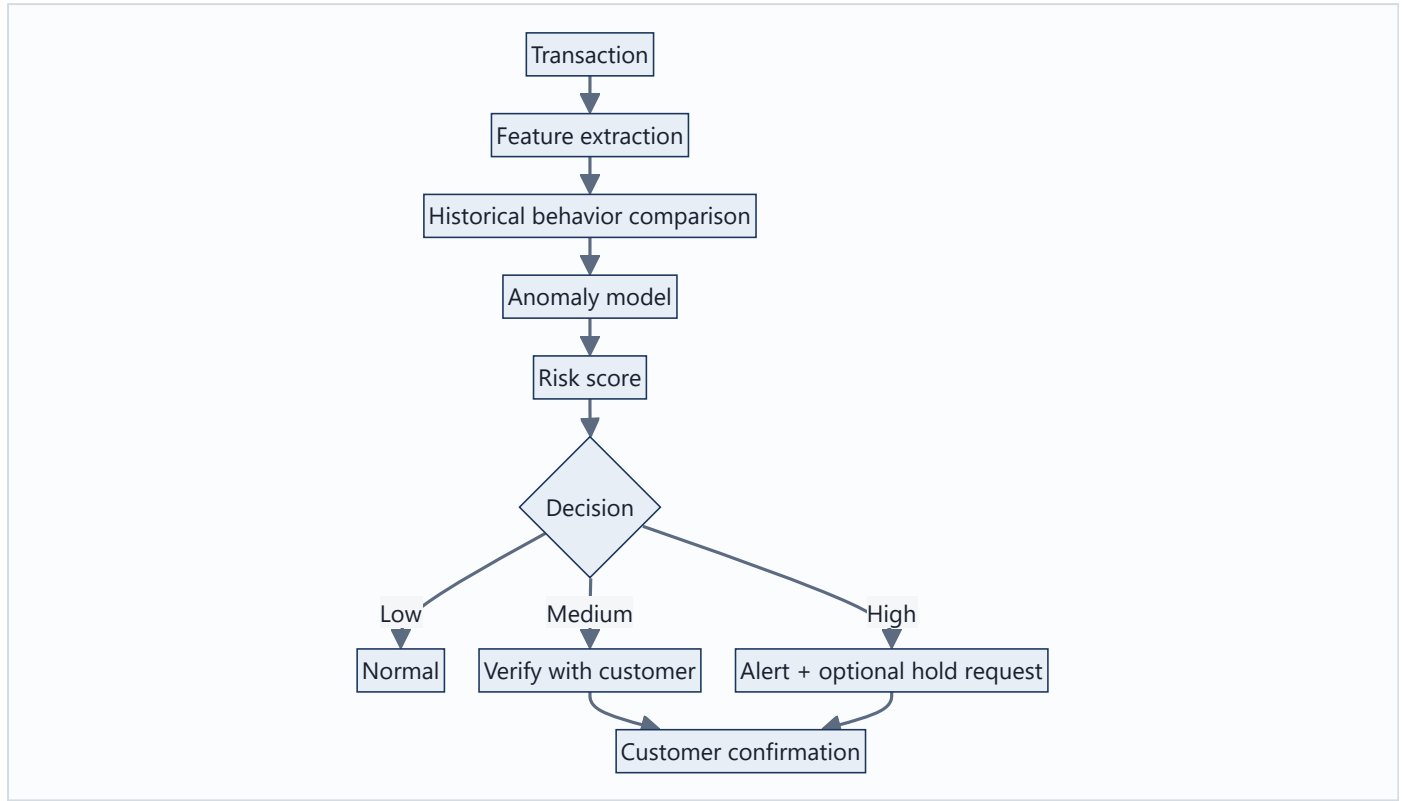
DIAGRAM — Financial stress



8.6 Fraud and anomaly — Kavach

Isolation Forest plus threshold rules (new payee, amount z-score, hour, velocity) against a 90-day baseline. Output: Normal / Verify / Alert. The MVP does not auto-freeze accounts. It asks the customer and logs the event.

DIAGRAM — Fraud detection



8.7 Vivek policy

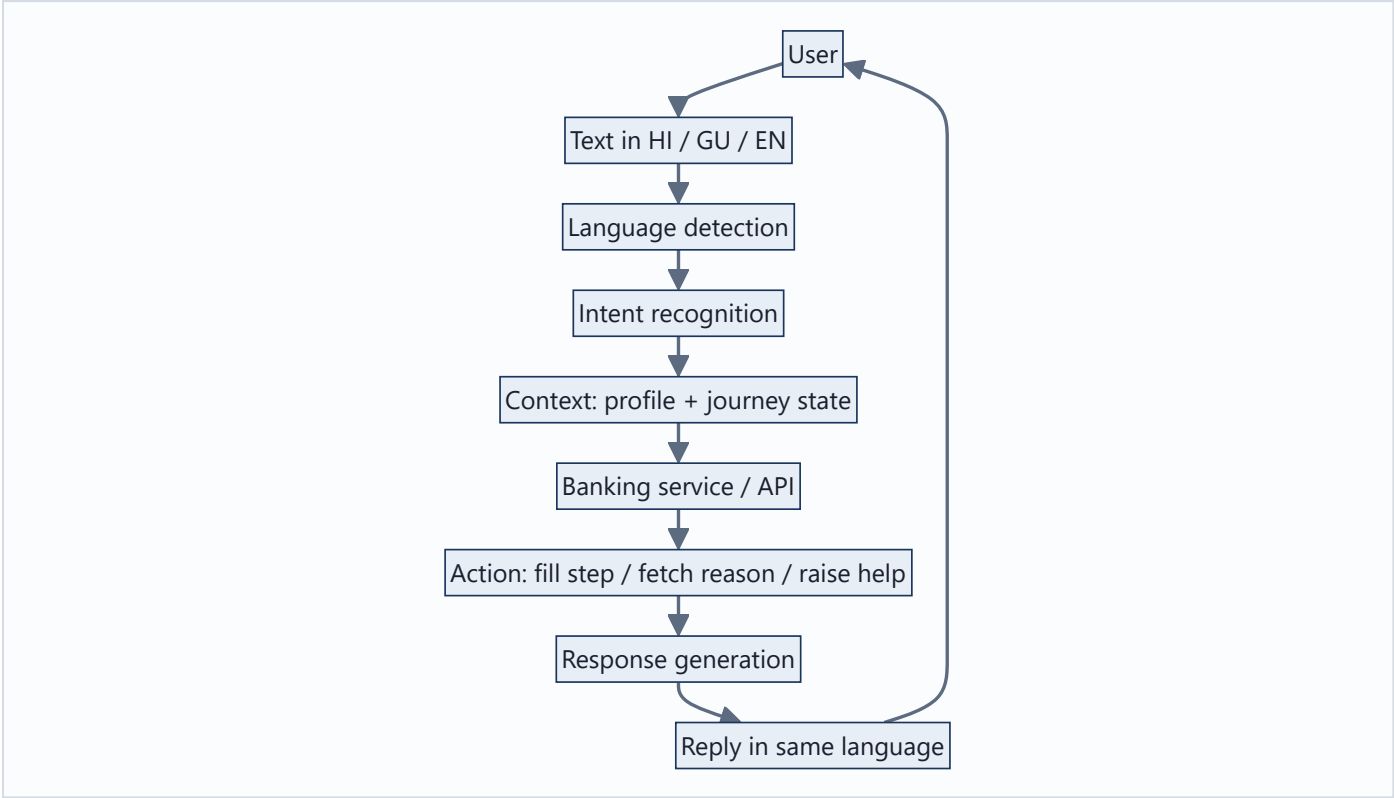
Deterministic gates, not a neural net. This is decision intelligence: every score maps to Recommend, Assist First, Suppress, or Verify.

9. Conversational AI — BhashaSahayak

Chat is a surface, not the product. It executes the same Vivek decision the dashboard shows. Text only in the MVP (Hindi, Gujarati, English). Voice is a later extension.

| Input  | User text, language hint, profile summary, allowed tools                                 |
|--------|------------------------------------------------------------------------------------------|
| Method | Language detection + intent + LLM with constrained tool calling; regex / langid fallback |
| Tools  | get_profile, get_recommendation, get_reasons, start_loan_step, confirm_txn               |
| Rule   | The model may translate reasons. It may not invent balances or reason codes.             |

DIAGRAM — Conversational flow

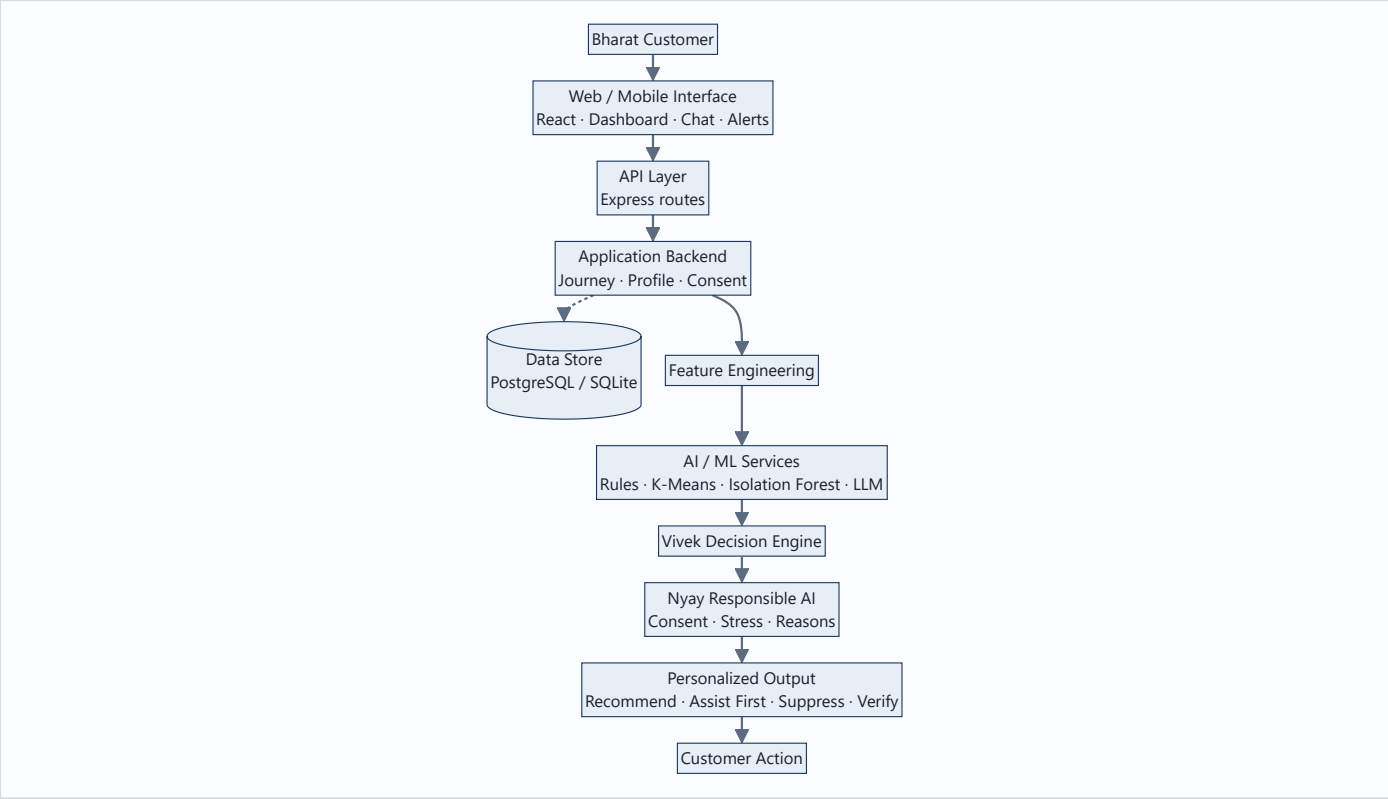


KYC in chat is *guidance* (checklist, plain-language steps). It does not replace CKYC or video KYC.

# 10. System Architecture

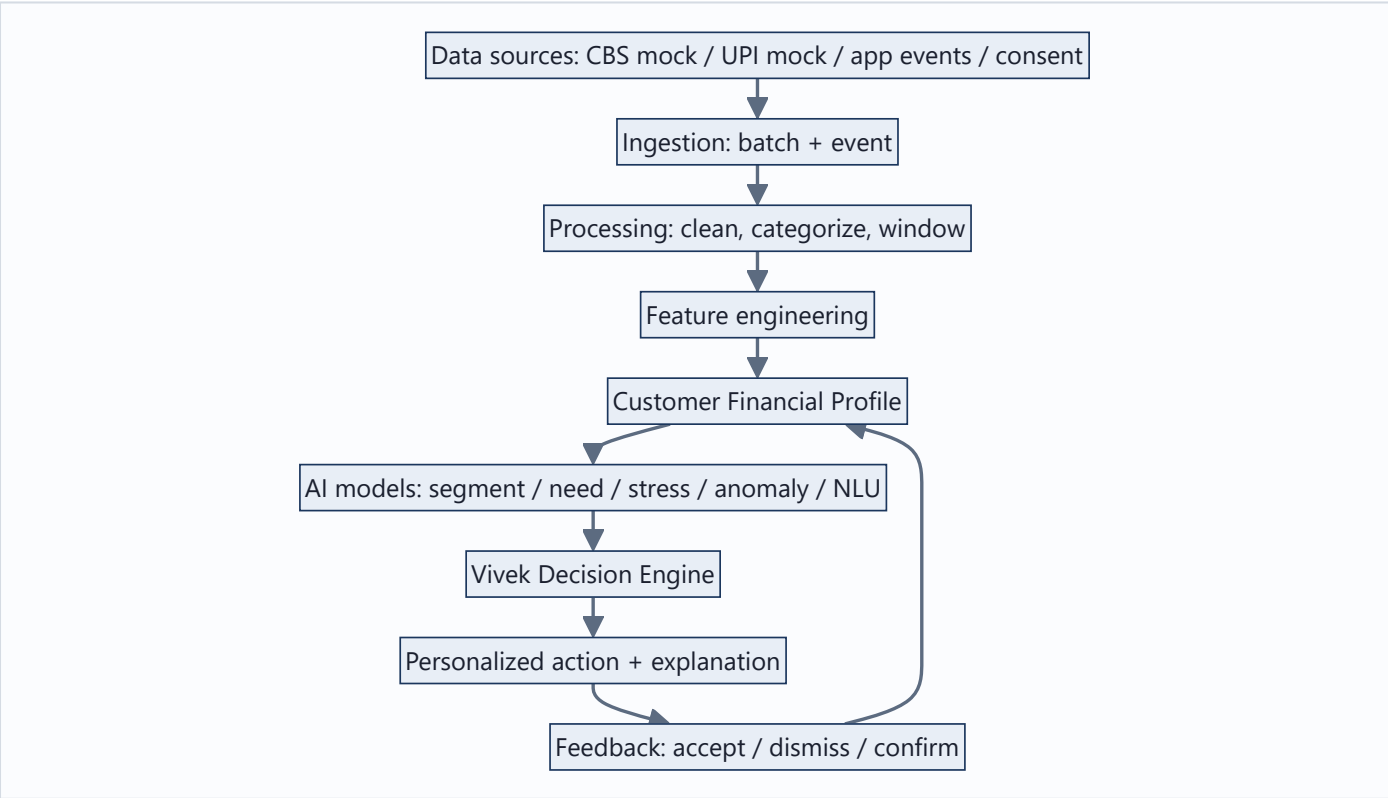
Customer → Web / Mobile → API → Application backend → Feature engineering → AI / ML → Vivek → Nyay → Personalized output → Customer action → Feedback.

DIAGRAM — High-level system



The data store is read and written by the backend. Intelligence is a downward chain. Output returns to the customer as a card, a chat reply, an empty suppress state, or a verify prompt.

DIAGRAM — Data flow



Every model has an input from the profile or the current transaction, and a user-facing action. Voice is absent from MVP diagrams.

## 11. Data Architecture

Data sources → Ingestion → Processing → Feature engineering → AI / ML → Storage → Output.

| Stage      | MVP                                               | Production                    |
|------------|---------------------------------------------------|-------------------------------|
| Sources    | Seed JSON/SQL ledgers, product catalogue, consent | CDC from CBS, UPI, app events |
| Ingestion  | Application boot and REST                         | Event pipeline                |
| Processing | Category mapper, monthly windows                  | Stream + batch                |
| Features   | Materialized on <code>financial_profiles</code>   | Feature store                 |
| Storage    | SQLite or PostgreSQL                              | Bank data platform            |
| Output     | Recommendations, alerts, chat, audit              | Same contract, more channels  |

**Sensitive.** Masked account numbers, amounts, salary, EMI, identifiers, chat text.

**Requires consent.** Transaction analysis, personalization, model improvement (separate purpose), chat-history retention.

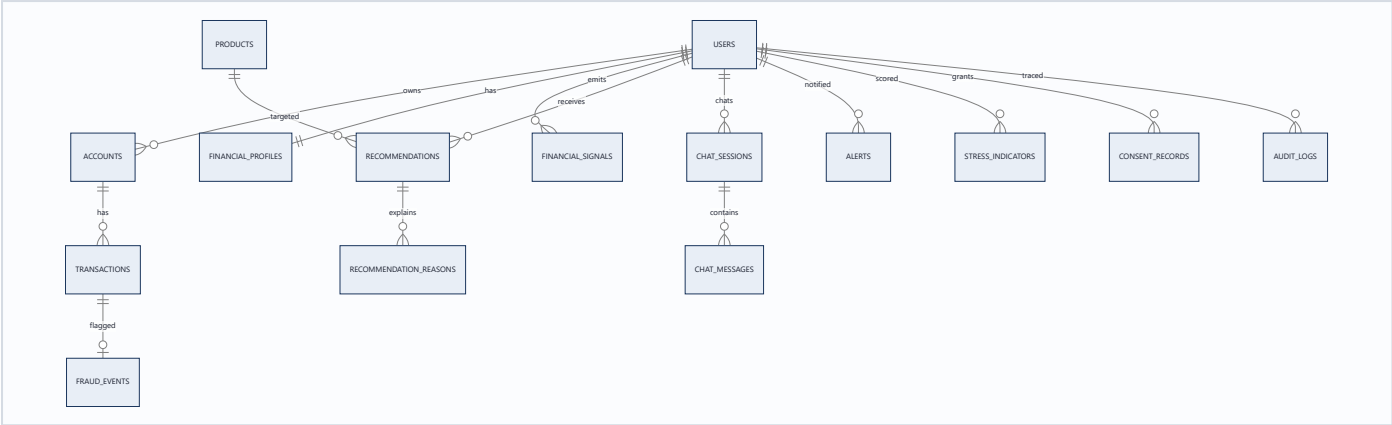
**Minimize.** Full PAN / Aadhaar are never stored in the MVP (production: CKYC vault). No raw statements beyond the feature window. Tokens and last-4, not full credentials.

## 12. Database Architecture

Entities exist only to support the features above.

| Entity                        | Purpose                           | Key fields                                           |
|-------------------------------|-----------------------------------|------------------------------------------------------|
| users                         | Identity, language, literacy mode | id, name, locale, channel                            |
| accounts                      | Deposit / loan / card shells      | type, masked_no, status                              |
| transactions                  | Ledger                            | ts, amount, type, category, channel                  |
| financial_profiles            | Latest intelligence snapshot      | income_est, emi_load, surplus, segment, stress_score |
| financial_signals             | Detected life-stage events        | signal_type, confidence, evidence                    |
| products                      | Catalogue                         | family, eligibility, suitability tags                |
| recommendations               | Issued decisions                  | action, score, ts                                    |
| recommendation_reasons        | Explainability                    | code, text in HI / EN / GU                           |
| chat_sessions / chat_messages | Assistant                         | locale, content, tool_trace                          |
| alerts                        | Stress / fraud / insight          | kind, severity, status                               |
| fraud_events                  | Anomaly cases                     | score, decision, customer_response                   |
| stress_indicators             | Score components                  | indicator, value                                     |
| consent_records               | Purpose-level consent             | purpose, status, version                             |
| audit_logs                    | Who / what / why                  | actor, action, entity, ts                            |

DIAGRAM — Entity relationships

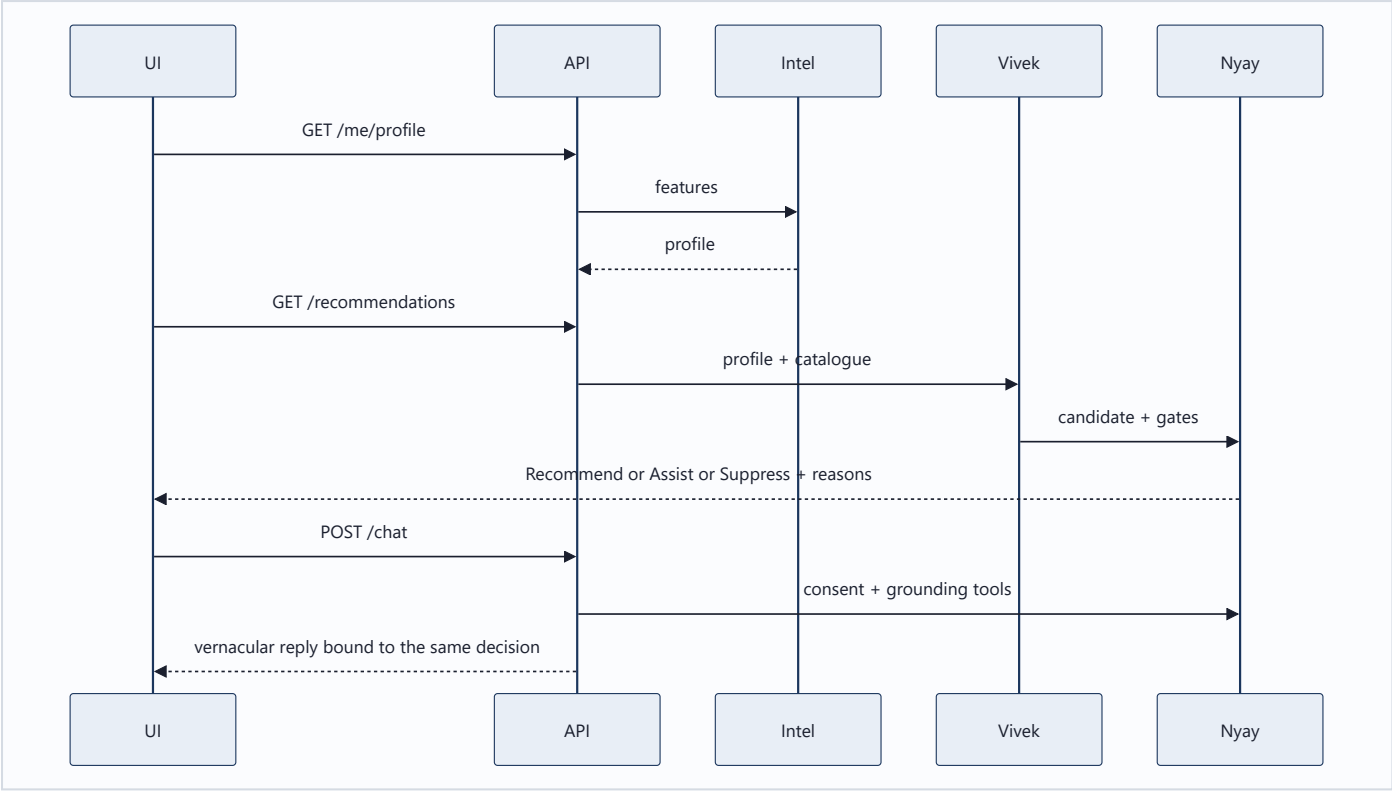


13. API Architecture

Base path /api/v1. Hackathon authentication is a session JWT. Personalization routes require a valid consent purpose.

| Method    | Endpoint                      | Purpose                 | Input                     | Output                       |
|-----------|-------------------------------|-------------------------|---------------------------|------------------------------|
| POST      | /auth/login                   | Sign in as a persona    | {personaId}               | {token, user}                |
| GET       | /me                           | Current user            | —                         | user + locale                |
| GET       | /me/profile                   | Financial profile       | —                         | profile + signals            |
| GET       | /me/transactions              | Masked ledger           | from, to                  | txn[]                        |
| POST      | /intelligence/refresh         | Recompute features      | —                         | profile                      |
| GET       | /recommendations              | Vivek decision set      | —                         | {action, items[], reasons[]} |
| GET       | /recommendations/:id/why      | Explanation             | id                        | reason cards                 |
| POST      | /recommendations/:id/feedback | Accept or dismiss       | {choice}                  | ok                           |
| POST      | /chat                         | Vernacular turn         | {text, locale, sessionId} | {reply, cards, locale}       |
| GET       | /wellness                     | Stress band and actions | —                         | score, band, actions         |
| GET       | /alerts                       | Open alerts             | —                         | alert[]                      |
| POST      | /fraud/:eventId/confirm       | Customer verify         | {recognized}              | updated event                |
| GET / PUT | /consent                      | Read or change purposes | {purpose, status}         | record                       |
| GET       | /audit/me                     | Customer-visible log    | —                         | events[]                     |

DIAGRAM — Recommendation and chat sequence



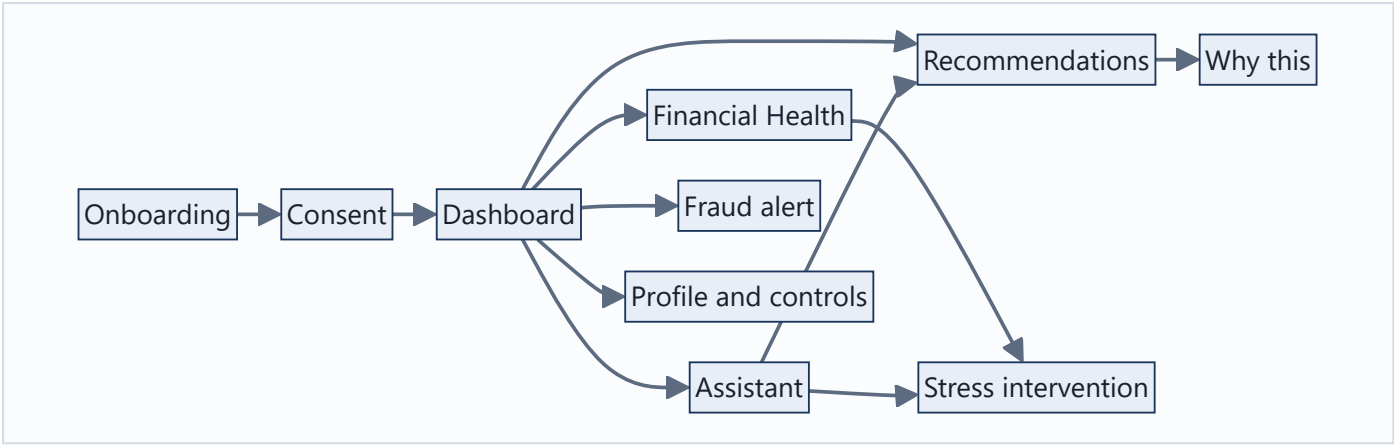
14. Frontend and UX Architecture

Literacy-first: large type, one primary action, persistent language toggle, rupee amounts spoken in words in chat, no deep hamburger maze.

| Screen              | Purpose                           | User actions                  | Backend                  |
|---------------------|-----------------------------------|-------------------------------|--------------------------|
| Onboarding          | Language and entry                | Pick HI / GU / EN             | /auth/login              |
| Consent             | Purpose-level opt-in              | Grant or skip personalization | PUT /consent             |
| Dashboard           | "Your money, understood"          | Open insight or chat          | /me/profile              |
| Financial Health    | Stress and buffers                | Read or start help            | /wellness                |
| Recommendations     | Honest products or empty state    | View, start, dismiss          | /recommendations         |
| Why this?           | Reasons and "what we did not use" | Trust or contest              | /recommendations/:id/why |
| Assistant           | Vernacular journey                | Type or tap chips             | /chat                    |
| Stress intervention | Empathy; no loan banners          | Ask help or talk to human     | /wellness                |
| Fraud alert         | Verify a transaction              | Yes / No                      | /fraud/:id/confirm       |
| Controls            | Language, consent, pause offers   | Change preferences            | /consent                 |

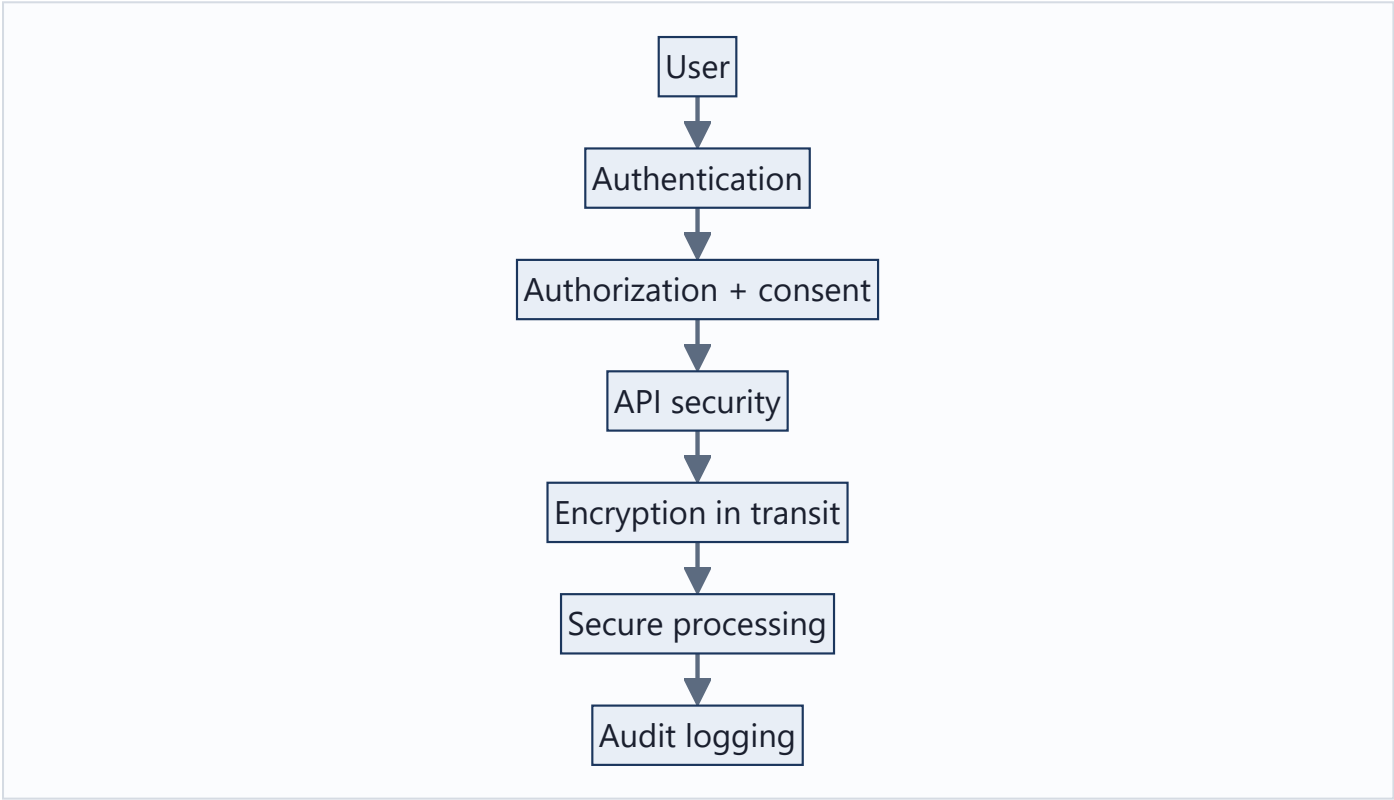
An empty recommendations state is a first-class screen, not an error.

DIAGRAM — Navigation



15. Security Architecture

DIAGRAM — Security path



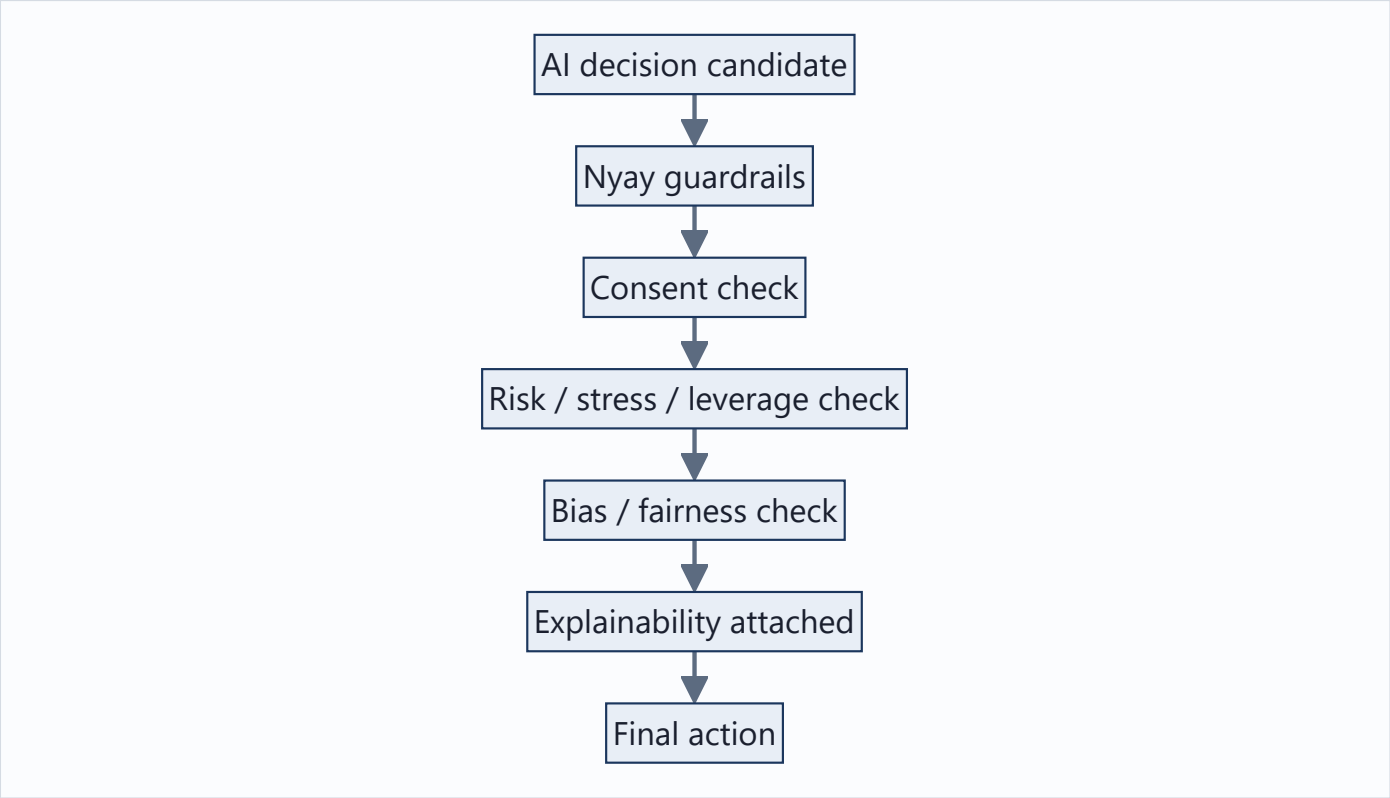
| Control        | Hackathon MVP             | Production banking                |
|----------------|---------------------------|-----------------------------------|
| Authentication | Demo personas + JWT       | Bank IdP, 2FA, device binding     |
| Authorization  | User can see only self    | Fine-grained roles, maker-checker |
| API            | Local CORS, basic headers | mTLS, WAF, rate limits            |
| Encryption     | HTTPS when deployed       | TLS 1.2+, field encryption, HSM   |
| Secrets        | .env not committed        | Vault and rotation                |
| Data           | Masked accounts in UI     | Tokenization, DLP                 |
| Audit          | audit_logs table          | Immutable store, SIEM             |

Hackathon security is demo hygiene. Formal information-security audit is not claimed.

## 16. Privacy and Responsible AI

Nyay is a runtime layer, not a policy appendix.

DIAGRAM — Nyay guardrails



Nyay enforces purpose-level consent; data minimization; reason codes; no countdown or “last chance” credit nudges; stress suppress; human escalation (talk\_to\_bank); append-only audit; customer pause-personalization; a separate purpose for model improvement.

Bias monitoring (later production): compare recommend-rates across language and income bands; alert if vernacular users are over-offered high-cost credit.

### Privacy and regulatory readiness

The architecture is designed with regulatory readiness in mind. Production requires formal validation.

| Theme               | Treatment                                                                               |
|---------------------|-----------------------------------------------------------------------------------------|
| DPDP                | Lawful purpose, vernacular notice, withdraw, minimize, access to “what you used”        |
| Purpose limitation  | Analyze transactions ≠ share with insurers ≠ train foundation models — separate toggles |
| Data minimization   | Features over raw dumps; retention windows                                              |
| Auditability        | Recommendation, consent, and chat tool traces                                           |
| Localization        | Production stores and processes as required for bank data; MVP uses a local database    |
| Lending conduct     | No hidden charges in UI; suitability; no credit increase when stressed                  |
| Responsible nudging | Suppress path is documented and visible                                                 |

This document does not claim that the application is fully RBI compliant or DPDP certified.

## 17. Explainability and User Trust

Every issued decision stores reason codes in Hindi, English, and Gujarati.

The Why-this screen lists why a product was recommended **and** what was not used (for example, no invented bureau score). Suppress is explained in the same vocabulary: “we are not offering new credit right now.”

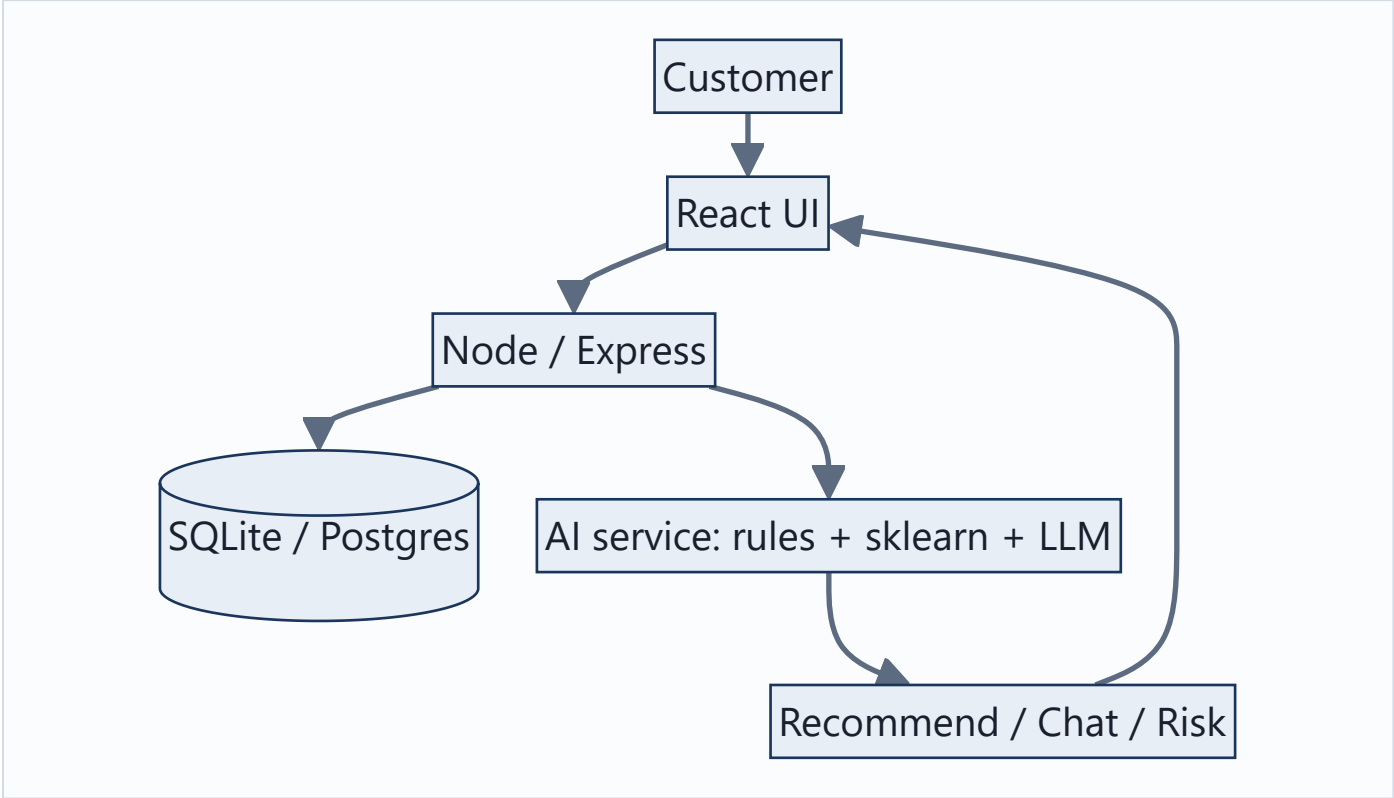
The LLM may translate a stored reason. It may not author a new one.

Customers can withdraw a purpose, pause personalization, and still use a simple account view. They can request a human. When the system is wrong, dismiss, confirm, and escalate are first-class actions; production adds model-risk review and appeal.

## 18. MVP Implementation

What the prototype actually does, on sample ledgers.

DIAGRAM — MVP runtime



| Capability                                                                                                | Status                                             |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Screens: onboarding, consent, dashboard, health, recommendations, why-this, chat, stress, fraud, controls | Implemented in MVP                                 |
| Express APIs listed above                                                                                 | Implemented                                        |
| Three persona ledgers                                                                                     | Sample data                                        |
| Feature jobs and life-stage signals                                                                       | Implemented on sample data                         |
| Vivek gates and reason codes                                                                              | Implemented                                        |
| Vernacular chat with tool calling                                                                         | Implemented, with scripted fallback                |
| Isolation Forest                                                                                          | Implemented or pre-scored if Python is unavailable |
| CBS / UPI / CKYC / bureau / SMS                                                                           | Mocked adapters                                    |
| Voice / video KYC                                                                                         | Not in MVP                                         |
| Compliance certification                                                                                  | Not claimed                                        |

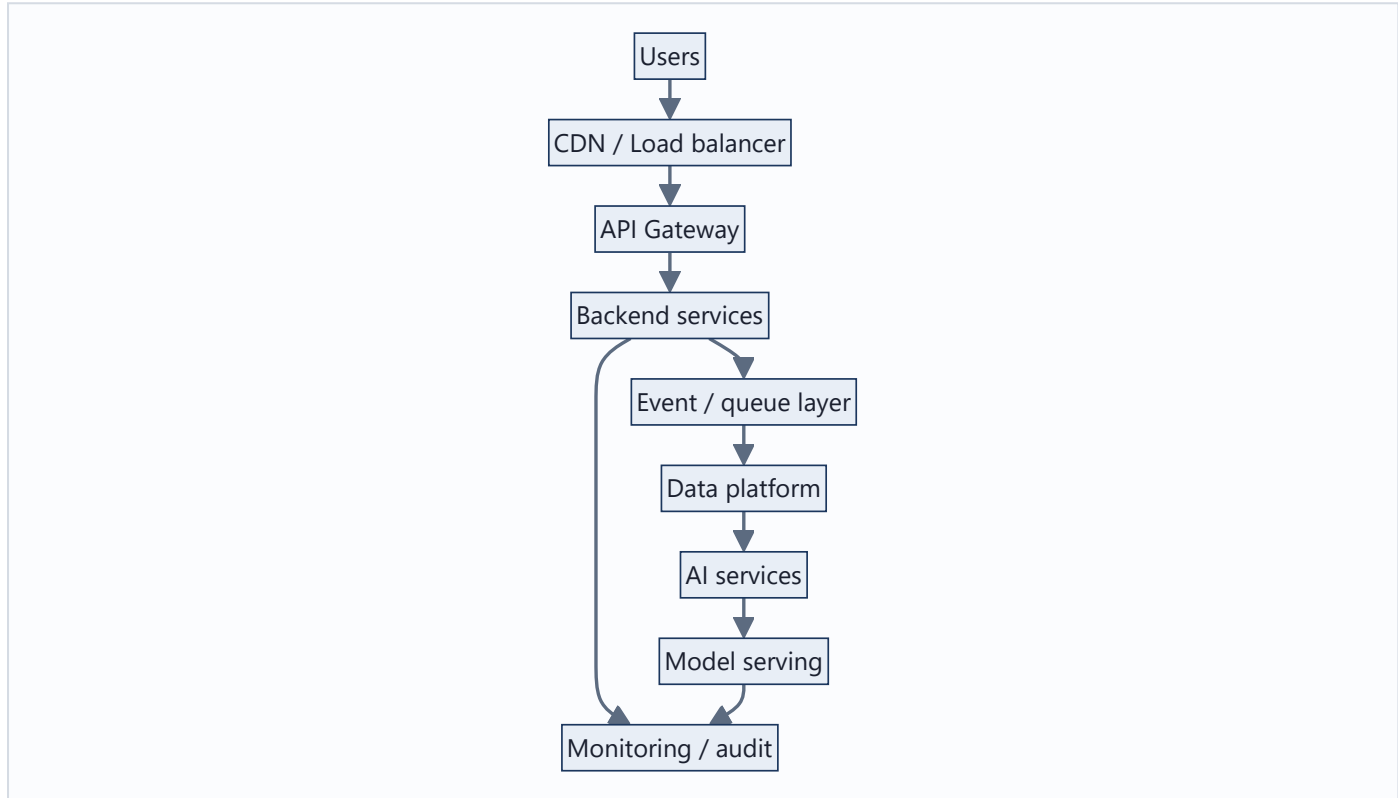
A missed-EMI change on the sample ledger flips Vivek from Recommend to Suppress. That is the implementation proof that the decision is computed, not hardcoded per persona name.

## 19. Technology Stack

| Layer    | MVP choice                                           | Why                              |
|----------|------------------------------------------------------|----------------------------------|
| Frontend | React + Vite                                         | Fast, screen-complete UI         |
| Backend  | Node / Express                                       | Journeys and APIs in one service |
| Database | SQLite or PostgreSQL                                 | Matches the entity model         |
| AI       | Rules + sklearn Isolation Forest / K-Means + LLM API | Explainable and appropriate      |
| i18n     | JSON strings + LLM replies                           | Hindi, Gujarati, English         |
| Infra    | Local or single VM                                   | Honest scope                     |

## 20. MVP versus Production

DIAGRAM — Production scale path



| Layer         | Hackathon MVP                | Production extension                |
|---------------|------------------------------|-------------------------------------|
| Edge          | Local Vite / simple host     | CDN, WAF, load balancer             |
| API           | Express monolith             | Gateway, versioned services         |
| Async         | In-request                   | Queue for transaction events        |
| Data          | Single database + seed files | Lake and feature store              |
| Models        | Rules + sklearn + LLM API    | Registry, A/B, drift monitors       |
| Observability | Request logs + audit table   | SIEM, model monitors                |
| Channels      | App prototype                | App, RM workstation, contact centre |

The decision contract stays the same: profile in, Vivek action and reasons out. Mocks become adapters.

## 21. Limitations

- Intelligence quality depends on the sample (or, in production, consented) ledger — not on a live bureau score.

- CBS, UPI, CKYC, and notifications are mocked.
  - LLM replies can vary; money figures must come from tools, not from free text.
  - Isolation Forest is not a full bank fraud platform.
  - Hindi and Gujarati do not cover all Bharat languages.
  - Bias monitors and calibrated stress probabilities are not in the MVP.
  - No formal compliance attestation.
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## 22. Future Scope

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Logical extensions of the same layer:

- Voice in additional vernaculars.
  - Live CBS / UPI / CKYC adapters after bank integration.
  - Calibrated stress models and collections playbooks.
  - Branch and relationship-manager workstation.
  - Production fairness monitors and model-risk governance.
  - Formal privacy and regulatory validation.
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## 23. Conclusion

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ARTHIX is an intelligent banking layer for Bharat. It understands consented financial context, personalizes the journey in the customer's language, protects people under stress, verifies unusual activity with the customer, and recommends a product only when it creates genuine value.

The same engine can recommend a suitable SIP to Rahul, help Meena finish KYC in Gujarati, and refuse to sell a loan to Kamala. That is the product: hyper-personalization with the discipline to stop.